Abstract

Contrary to the ideas voiced by proponents of the "Cultural Backlash" theory, this essay argues that LGBT rights did not have an effect on populist vote share in Europe. Using a staggered difference-in-differences, modern event-study estimators and robust controls, I find no statistically significant electoral backlash against the introduction of civil union as a right. These results contrast with cultural backlash theories and remain robust to multiple specifications and placebo tests.

I. Introduction

Contrary to the ideas voiced by Inglehart and Norris (2016), this essay argues that LGBT rights did not have an effect on populist vote share in Europe. Particularly, this essay looks at the effect of the introduction of civil union as a right on the vote share of the populist right of 33 European countries.

Most democracies around the world, including Europe, have experienced a rapid rise in the vote share of populist candidates, particularly populist right candidates (see Figure 1). At the same time, most European countries have progressively introduced various LGBT rights (see Figure 2). This seems to support the cultural changes thesis, voiced by Inglehart and Norris (2016).

Figure 1. Populist Vote Share, 1980-2020, n=33

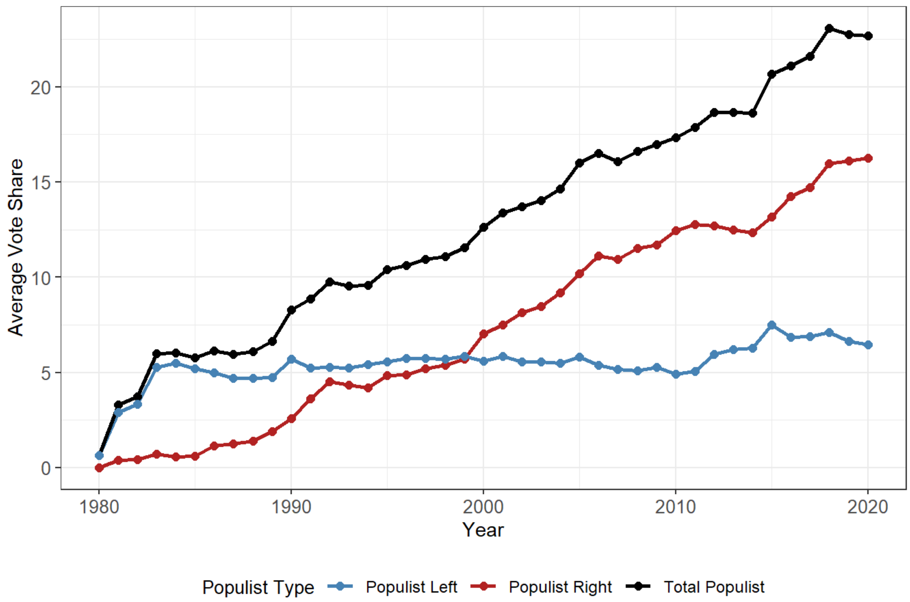
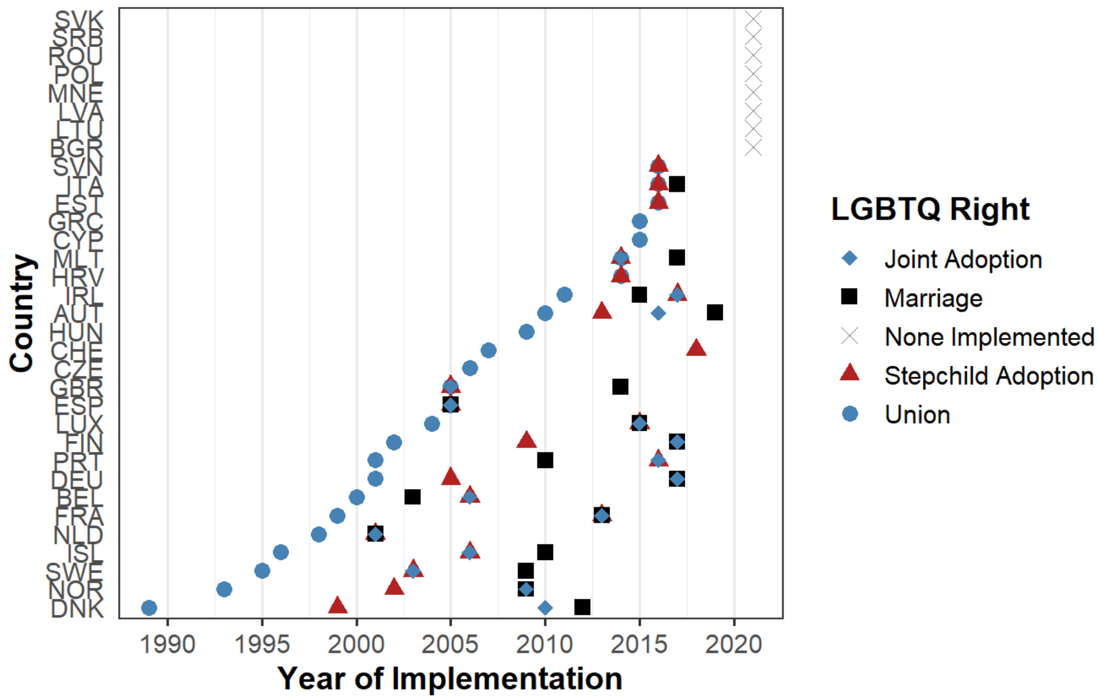


Figure 2. Event timeline chart of LGBT rights implementation by country

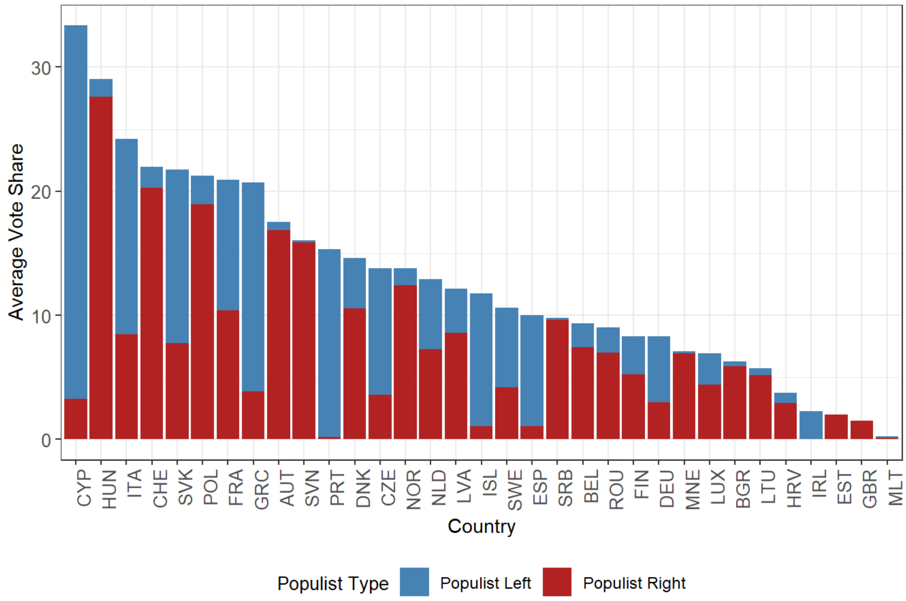


Nonetheless, European countries over this time period have also experienced a multiplicity of events potentially influencing populist vote share. Various of them have been associated with populist vote share, such as economic insecurity (Funke, Schularick and Trebesch, 2016; Autor *et al.*, 2020; Guiso *et al.*, 2024), green policies (Colantone *et al.*, 2024), and immigration (Pupaza and Wehner, 2023; Alrababah *et al.*, 2024). These render any naïve cross-country regression attempting to better understand what caused this increase in vote share fundamentally biased, even when trying to control for *everything*. This motivates the use of a suitable identification strategy that can help isolate the effect of the introduction of LGBT rights on populist vote share.

II. Descriptive Statistics

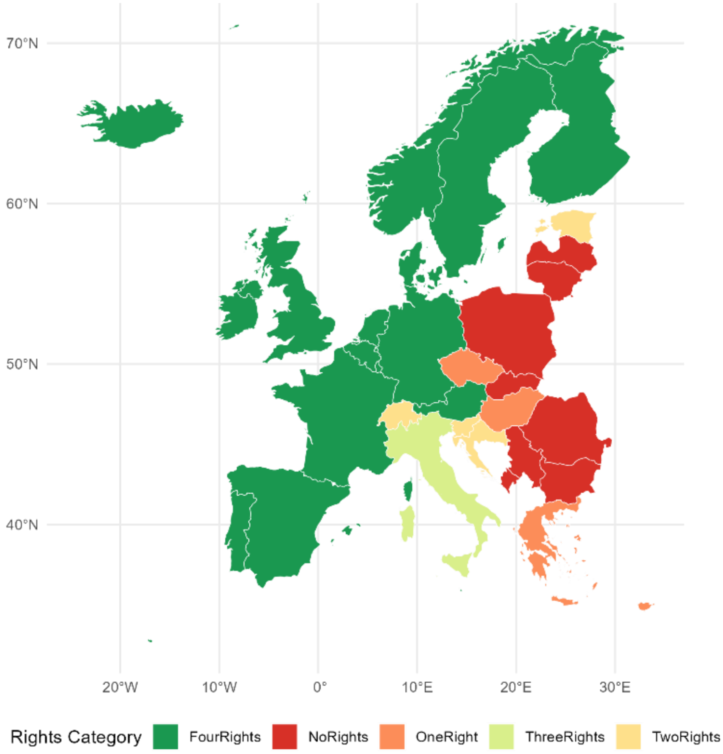
Our panel data has 33 European countries observed between 1980-2020, after dropping 1979 and 2021 due to a lack of observations in either year. This panel has data on the left and right populist vote share attained in each parliamentary/legislative election for these countries over this time period, recovered from Timbro. See figures 1 and 3 for some descriptive statistics on populist vote shares. In Figure 3 we can see the average populist vote share of all the countries in our dataset. We can see some countries have either mostly left- or right-wing populists, but the general trend is that populist right candidates have been on the rise since the 1990s.

Figure 3. Average populist vote share by country and left-right spectrum



The dataset also includes variables for four different LGBT rights (civil union, stepchild adoption, joint adoption and same-sex marriage) and the year they were adopted, if so, by each country (see Figure 2). Notably, no European country has ever introduced and then rescinded any of these LGBT rights. See Figure 4 for map on implementation of LGBT rights.

Figure 4. Map of LGBT rights implementation



The binary civil union variable used in the regressions is constructed in a way that it equals one in the first post-law election year (or if introduced the same year), and thereafter; zero otherwise. This is done to avoid introducing bias in the estimate, as in some cases the right will be introduced various years before an election and, given vote shares cannot change outside elections, it is unreasonable to run a regression where the effect will be mechanically zero, as this would produce a downward bias on the estimate.

Control Variables

The dataset includes data from Eurostat on: population over 65 years old as a share of total population, which Inglehart and Norris (2016) identified as a predictor of favouring populist candidates; immigration percent as a share of total population, which Pupaza and Wehner (2023) and Alrababah *et al.* (2024) both identify as increasing populist right vote share; GDP growth, which may capture the financial crisis aspect identified by Funke, Schularick and Trebesch (2016); unemployment rate (as identified by Guiso *et al.*, 2024), which should indicate the differences within Europe with regards to economic insecurity; and import penetration as a share of GDP, which may capture rising Chinese import competition in particular countries (Autor *et al.*, 2020). I have merged turnout data for each of the parliamentary elections covered, which will be interacted with unemployment to use Guiso *et al.*'s specification. Some descriptive statistics on the variables of interest can be found in Table 1.

The rationale for including these variables as controls in my estimations is that these will improve the civil union point estimates, while reducing the chances of omitted variable bias. For example, unemployment may influence when civil union is introduced as a right because under high unemployment the policymakers' agenda will likely prioritise economic policies over LGBT ones, but unemployment should also influence populist vote share, as identified by Guiso *et al.* (2024). By controlling for unemployment, and the other variables mentioned, we are reducing the risk from omitted variable bias, as many of these will confound with treatment *and* outcome.

III. Research Design

Choice of identification strategy

This essay will employ a staggered difference-in-differences design. The baseline regression model estimated is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{g \neq -1} \beta_g \cdot D_{it}^{(g)} + \boldsymbol{X}'_{it}\boldsymbol{\gamma} + \varepsilon_{it}\,, \tag{1}$$

where:

$$D_{it}^{(g)} = \mathbf{1}\{t - G_i = g\}$$

Table 1. Descriptive Statistics of Main Variables

Variable	N	Mean	Median	SD	Min	Max	Countries	Years
Growth (% of GDP)	857	5.68	5.08	7.50	-48.83	47.93	32	41
Immigration (%)	774	0.93	0.72	0.85	0.00	5.74	30	30
Import (% of GDP)	570	13.55	11.16	6.69	0.00	38.36	27	22
Population >65 (%)	1,268	15.03	14.94	2.93	7.98	23.24	33	41
Populist left vote share (%)	1,353	5.41	1.30	8.40	0.00	44.90	33	41
Populist right vote share (%)	1,353	7.38	2.60	10.72	0.00	69.50	33	41
Turnout (%)	1,162	73.08	74.91	13.92	31.95	98.87	33	41
Unemployment (%)	855	8.37	7.46	4.33	1.54	27.49	30	38
Civil union	1,353	0.26	0.00	0.44	0.00	1.00	33	41

The outcome (Y_{it}) is the populist right vote share of country i at year t, while the treatment variable, $D_{it}^{(g)}$, is a series of relative time-indicators (event-time dummies) that equal one in the g^{th} year relative to the first year G_i in which country i adopts civil-union as a right *and* there's an election (and all years thereafter); or zero otherwise. The coefficients β_g will capture the dynamic treatment effects at each relative period, accounting for treatment timing heterogeneity. $\boldsymbol{X}'_{it}\boldsymbol{\gamma}$ is a vector of time-varying controls, specified in the previous section. α_i represents unit-specific fixed effects, while λ_t represents time-specific fixed effects. These absorb country-level baselines and common shocks, focusing identification on within-country changes.

The rationale for using populist right vote share as my dependent variable is that this is the most direct expression of the recent rise of populism in Europe, as populist left vote share has remained fairly constant throughout the period that LGBT rights were introduced (see Figure 1). While studying the progression of populist left vote share may also prove interesting in other contexts, this would not be an adequate form of testing the cultural backlash thesis, as under such theory it is unlikely voters would vote more for the socially-progressive left as a reactionary measure. Using total populist vote share as the outcome variable should provide similar estimates to populist right, but would have some noise from changes in populist left vote share particularly after 2010, when the Eurozone crisis occurred, and parties like Syriza in Greece or Podemos in Spain rose. This may lead to the misinterpretation that the increased vote share of these parties is due to the introduction of civil union.

Civil union is a particularly good measure of the effect of LGBT rights on populist vote share for a variety of reasons. First, civil union is always the first right introduced by a country, even if in some cases it is introduced alongside other rights (see Figure 2). Second, while only 17 countries had introduced same-sex marriage as of 2020, 25 had introduced civil union (see Figure 2), giving us additional data points to work with. Third, and most importantly, civil union is sufficiently visible and relevant that, if the Cultural Backlash theory is true, voters should react to it.

Alternative rights, such as stepchild adoption, joint adoption or marriage, are most often introduced around the same time, and always follow prior rights (i.e., civil union), so their effects are noisy and difficult to isolate. Additionally, while relevant in their own sense, stepchild and joint adoption were likely not as media-relevant, so their effect would be a lower-bound. Civil union, on the other hand, likely provides an upper-bound of the effect on populist vote share.

The reasons for using staggered diff-in-diff mostly come from the difficulty in using alternative identification strategies, such as regression discontinuity (RDD) and instrumental variables (IV). Both strategies present problems with regards to their identification assumptions, with RDD suffering from the lack of an operational continuous running variable on which to create a plausible bandwidth of quasi-randomness, and IV suffering from the lack of suitable instruments which do not have some open back-path through unobserved confounders with the outcome.

Staggered difference-in-differences, on the contrary, has a lot of upsides for this particular research question. Given LGBT right adoption is staggered through time, and that we have data for multiple countries over multiple years, we can leverage such variation and data to check whether the implementation of civil union had an effect on populist vote share, controlling for country and year-specific variation, as well as a set of time-varying controls.

Staggered difference-in-differences has seen a “revolution” in the past few years, with various papers showing that what TWFE diff-in-diff was doing in the “background” was not exactly what was once thought (Baker *et al.*, 2025). In short, TWFE event-study regressions pool all cohorts’ relative time indicators into a single set of coefficients, which under staggered adoption and heterogeneous effects yields “contaminated” estimates which make “forbidden comparisons” between early and late treated cohorts, with possibly counter-intuitive negative weights. To address these concerns, this essay will use Sun & Abraham's (2021) modified difference-in-differences estimator, which avoids this by estimating cohort-specific event-study effects and then aggregating them using only not-yet-treated and never-treated units as control.

The results obtained with "traditional" TWFE will be presented too, to illustrate the differences in estimations and to contribute to the debate on the "low power" of "modern" estimators (as Chiu et al., 2025 and Weiss, 2025 have shown). Callaway & Sant'Anna's (2021) and Liu, Wang and Xu's (LWX) (2024) own estimators will be tested and presented as well for robustness, given their slightly different approaches to solve the issue of "forbidden comparisons" in estimating heterogeneous treatment effects.

The main identification assumption for staggered difference-in-differences is parallel trends, which states that the treated units would have followed a similar trend to control units in the absence of civil union being introduced. In the words of potential outcomes, the potential outcome for treated units under control is expected to be the trend of control units in the post-period plus the pre-treatment differences. The main test to check if this assumption is met is through an event-study plot, which plots the evolution in differences between the treated and control units with regards to the outcome of interest, which in this case is the populist vote share of country *i*. While these plots do not give definitive proof over whether the assumption is met, they are the best tool available.

Additionally, another assumption of difference-in-differences is no anticipation (Baker *et al*., 2025). While event-study plots do provide some information on whether there is a significant violation of it through pre-treatment time coefficients, a way to check for no-anticipation is through timing placebos, which in essence are "fake treatment" dates. If we find effects before real treatment, this would suggest a violation of the no-anticipation assumption. Four years will be tested (two, three, four and five years before adoption), as these should encompass most election cycles. If indeed there was anticipation, we should see positive, statistically significant effects. Nonetheless, if anticipation only arises a few months prior to the policy being introduced, yearly data will not be able to identify it.

IV. Results

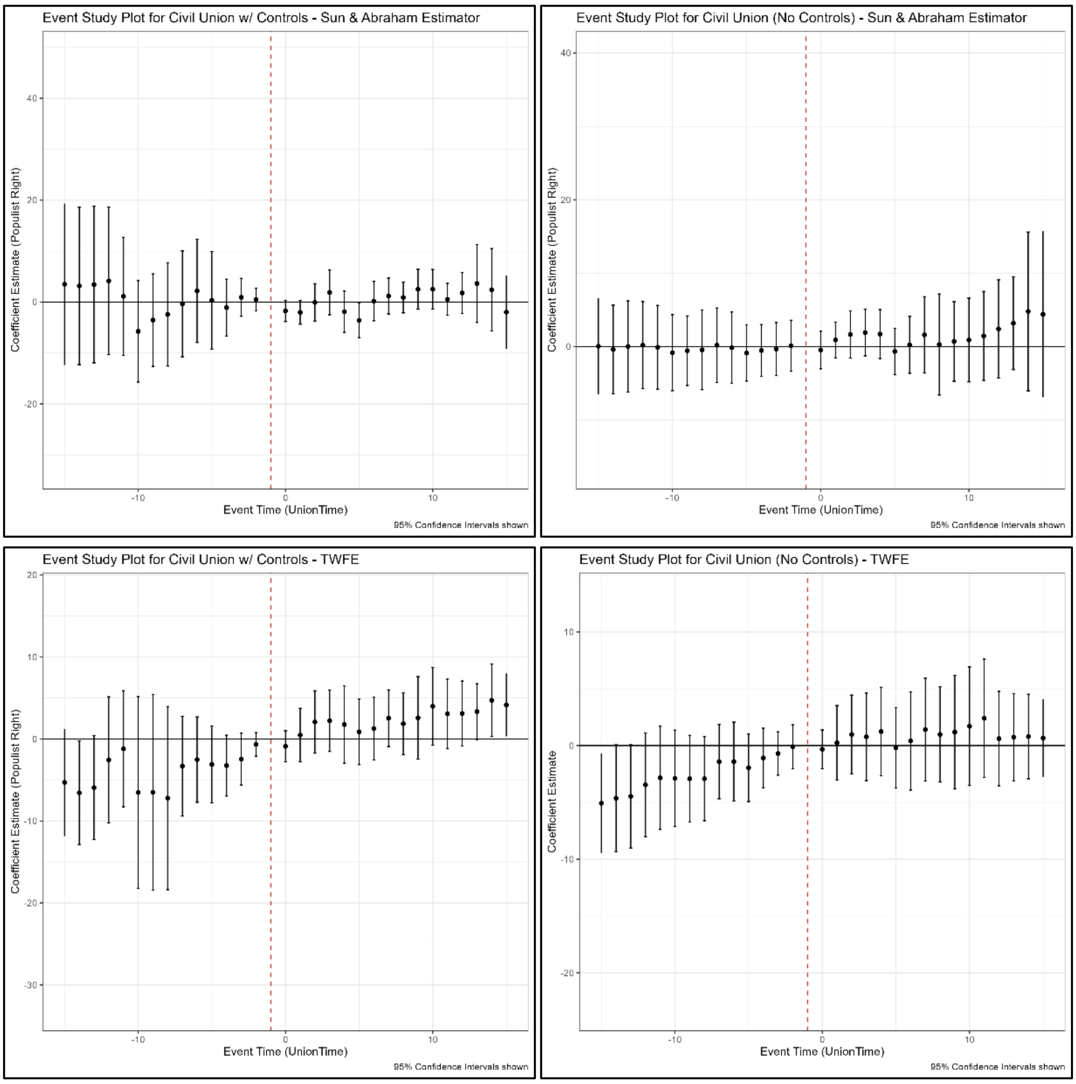
In Figure 5 we can see the main event study plots for our model ±15 years from the introduction of civil union as a right, with and without controls ($\boldsymbol{X}'_{it}\gamma$), and with both the Sun & Abraham estimator as well as "traditional" TWFE. The first notable result is that parallel trends seem to hold across different specifications and estimators, conditional on our set of covariates, with only one statistically significant deviation in the TWFE estimation without controls in the 15$^{\text{th}}$ year prior to the introduction of civil union.

The second most important result from these plots is that it seems the effect from introducing civil union on populist right vote share is not statistically significant at the 95% level in any year or specification used here, although the coefficient five years after treatment in the Sun & Abraham model with controls would result statistically significant at the 90% level, with a coefficient of -3.6 (p-value=0.05004). This would contradict the expectations from the Cultural Backlash theory, as it would signal that introducing civil union as a right diminishes the vote share of the populist right by 3.6-percentage-points. This may be due to increased political engagement of the LGBT community rewarding political action in their favour, or a normalisation (cancellation) of LGBT-communities (anti-LGBT groups).

If instead of looking at dynamic treatment effects we were interested in the aggregated effect, we would find similar results, as showcased in Table 2. These results are only statistically significant in our baseline model with no fixed effects or controls, increasing the populist right vote share by 5-percentage-points. Nonetheless, in the remaining specifications, no statistically significant effects are found at the 95 or 90% significance levels. The coefficient from the model with all controls indicates a 1.4-percentage-points increase in the vote share of the populist right after introducing civil union as a right, corresponding to an 8.8% of the mean populist right vote share in 2020, or 19.5% of the mean populist right vote share between 1980-2020 in our sample. Given the effect is not statistically significant at any conventional significance levels, we cannot interpret this effect as causal, as it may be due to the loss of sample that including covariates generates. The results still indicate that the effect from LGBT rights on populism is at least not as direct nor generalisable as indicated by Inglehart and Norris (2016). While the effect may indeed be significant and positive when looking at surveys, a subset of countries, or when using a different specification, it is not here.

To check whether the no-anticipation assumption is violated I conduct four timing placebo tests, which re-assign treatment dates in the pre-period (see Figure 6). No statistically significant results are found in any of the "fake treatment" periods. Again, while this is not proof that the no-anticipation assumption is met, it strengthens the argument that introducing civil union as a right did not increase the populist right's vote share.

Figure 5. Event Study Plots with Sun & Abraham (2021) and TWFE estimators



V. Limitations

The main limitation of this study comes from the potentially narrow scope of using a single right as treatment. Doing these same tests for different rights may provide different results, more consistent with the Cultural Backlash theory's expectations.

Another limitation of this study is that we cannot check whether there were policies or debate preceding the introduction of civil union as a right which minimised any political backlash from the right itself, violating the no-anticipation identification assumption. While in the event study plots and placebo timing tests we do not see any signs of anticipatory effects, there may be within-year anticipatory effects which we cannot observe in our yearly, vote-share data.

Table 2. Effect of Civil Union on Populist Right Vote Share - Aggregated

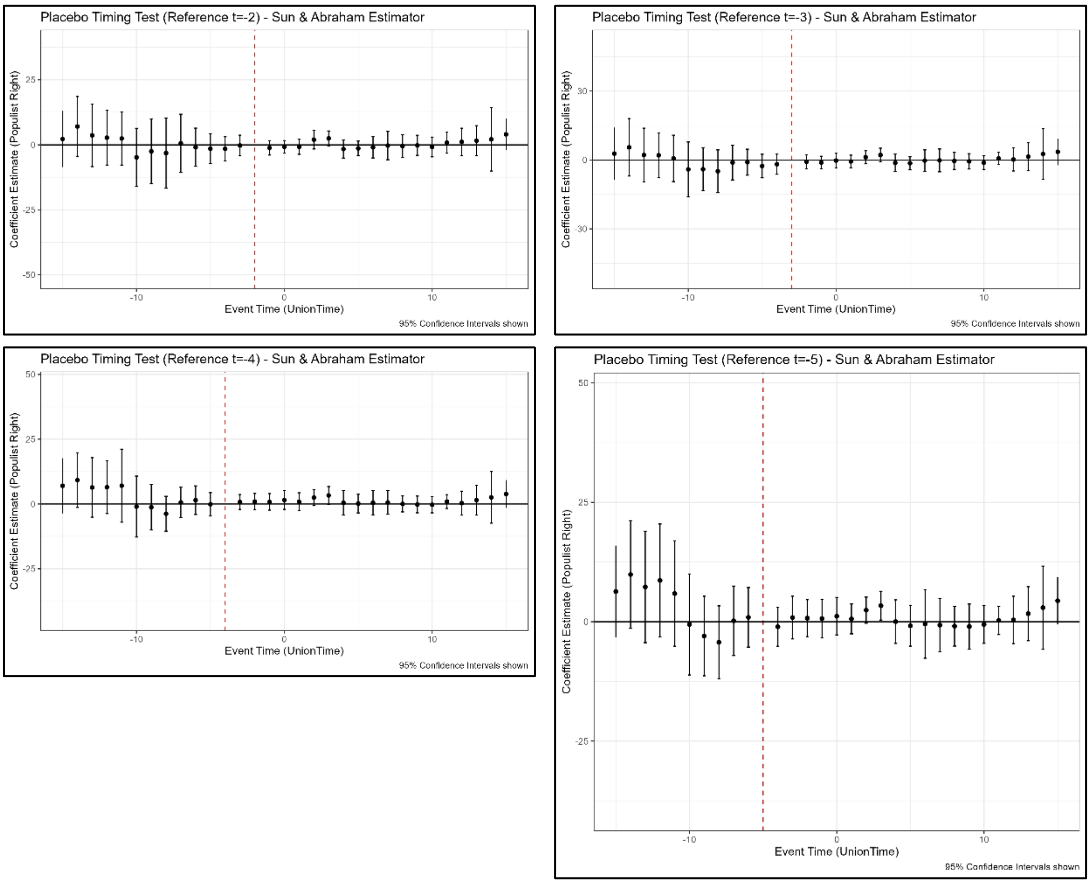
	Base	(1)	(2)	(3)	(4)	(5)
Civil Union	5.228*	-2.485	-1.551	-0.589	0.946	1.443
	(2.024)	(2.213)	(2.119)	(2.369)	(2.835)	(2.955)
Population age control	No	No	Yes	Yes	Yes	Yes
Immigration control	No	No	No	Yes	Yes	Yes
Growth-import controls	No	No	No	No	Yes	Yes
Guiso *et al.* controls	No	No	No	No	No	Yes
Country-FE:		X	X	X	X	X
Year-FE:		X	X	X	X	X
Observations	1353	1353	1268	765	523	522
R^2	0.589	0.595	0.598	0.731	0.811	0.818

Notes: Clustered standard errors are in parentheses (country). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The last column corresponds to the same specification as the full set of controls in the event study plots presented above.

Additionally, we also do not know which party introduced civil union as a right, which may explain why we do not find a statistically significant effect, as it may be that this is what determines whether there is "cultural backlash" or not. We also do not know the ideological alignment of the non-populist parties. It may be that non-populist parties are driven more to the "extremes" and also support anti-LGBT rights policies which "takes" from the populist right vote share. A way to test for this would be to test for the vote share of all parties identified with the right wing, rather than just populists.

Another significant limitation of the study comes from the treatment variable's operationalisation. This operationalisation implicitly assumes that the effect should be the same regardless of whether the next election following the right's introduction is four years away or just one. This issue may overestimate the effect of civil union on populist vote share, as it may indicate it had an effect that does not correspond to civil union legislation but some more recent event. No easily-implementable correction to this issue could be thought of in the preparation of the research design, and this should be addressed in further studies. A partial solution may be considering only countries

Figure 6. Placebo timing tests - Reference periods t = {-2, -3, -4, -5}



where the right was introduced at most a year before the election, although in this sample this was only the case for 11 of the 25 countries implementing civil union, which may generate some small-sample issues. Another alternative, even if imperfect, may be interacting the treatment with the time until the next election, or using a measure of how much the country's media talked about the policy.

This research design, while plausibly meeting all the relevant identification assumptions, would benefit from further tests checking for heterogeneity in the treatment effect. This may include interacting the civil union event-dummy with seemingly plausible factors determining whether a country reacts negatively to civil union being introduced as a right, such as average population age, historical religious affiliation or ethnic fractionalisation.

VI. Additional Tests

As additional tests, the results obtained from Callaway & Sant'Anna's (2021) $ATT(g,t)$ and LWX's (2024) interactive fixed effects counterfactual ($IFEct$) and matrix completion (MC) estimators will be presented in figures 7 and 8.

Figure 7. Callaway & Sant'Anna's (2021) $ATT(g,t)$ – Event Study & Group Aggregated Coefficients

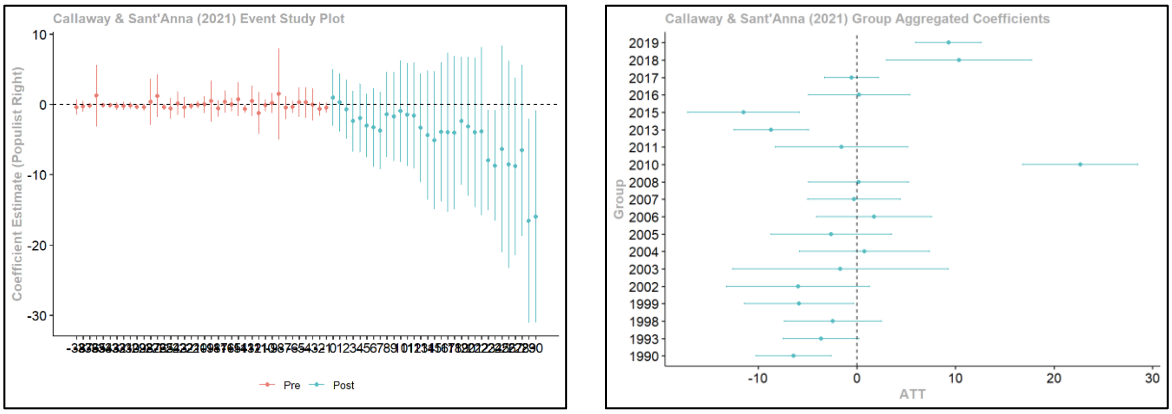
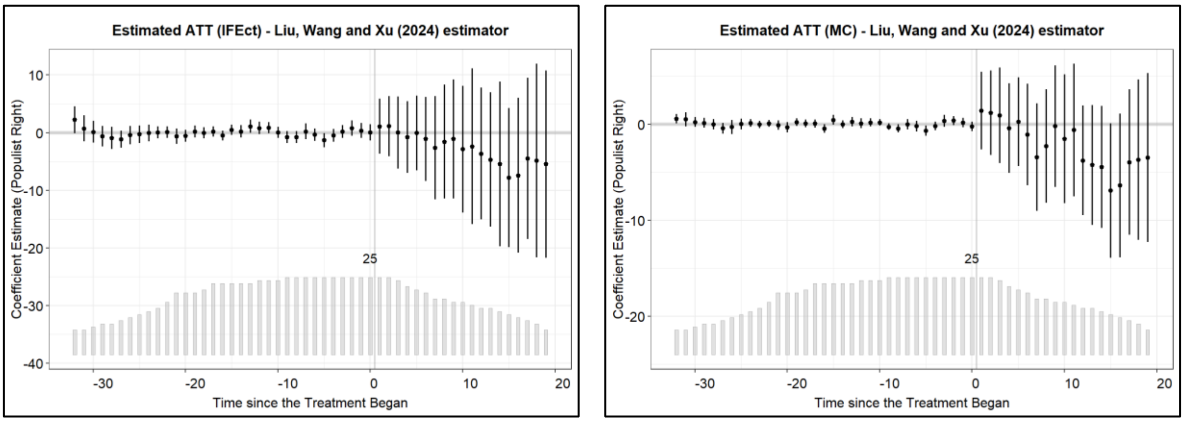


Figure 8. Liu, Wang and Xu's (2024) $IFEct$ & MC estimators



A "problem" with the Callaway & Sant'Anna (2021) estimator is that you cannot include time-varying covariates while estimating it, driving some differences with the main model's results. Nonetheless, in the results presented, parallel trends still seem to hold, and no statistically significant effects can be found. By looking at the group-aggregated coefficients we get an interesting finding, as it seems that those who implemented civil union as a right in 2010, 2018 and 2019 experienced an increase in their populist right vote share. Still, as these year-groups represent at most 2 countries, the sample is small enough that no substantive interpretation of the results is warranted.

LWX's (2024) $IFEct$ and MC estimators are a deviation from canonical difference-in-differences as they in essence estimate a control value post-treatment for the treated units (counterfactuals). This method is not comparing treated units with never treated or not-yet-treated units like the other estimators are. Their results strengthen the argument that civil union did not increase the populist right's vote share, as they still fail to find a statistically significant effect.

VII. Conclusion

Given the evidence presented here, it does not seem like the introduction of civil union as a legal right had a statistically significant and positive effect on the vote share of populist right candidates in Europe. This lack of an effect holds across multiple specifications and staggered difference-in-difference models. Previous studies, such as that by Inglehart and Norris (2016), which found statistically significant and positive effects, may have used low-quality survey data, misspecified their models, or lacked a suitable causal identification strategy. This essay, while still presenting significant limitations, points to a more nuanced reality than the Cultural Backlash theory seems to suggest, in which LGBT rights do not directly translate into higher populist right vote shares.

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